



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

Bootstrap Framework

JUNE, 2023



Bootstrap: Introduction

- Bootstrap is the **most popular HTML, CSS, JavaScript framework** for developing a **responsive** and **mobile friendly** website.
- It is **free** to download **front-end framework**.
- It can be utilized for **easier** and **quick web deployments**.
- It contains **HTML/CSS based design templates** for buttons, tables, typography, navigation, form, modals, images, carousels and many more.
- It provides facility to easily implement **responsive designs**.



Bootstrap: Introduction

- Bootstrap was developed by **Mark Otto** and **Jacob Thornton** at Twitter and released as an **open-source product** in August **2011 on GitHub**.

Why Bootstrap?

- **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- **Responsive features:** Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- **Mobile-first approach:** In Bootstrap 3, mobile-first styles are part of the core framework
- **Browser compatibility:** Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Edge, Safari, and Opera)



Bootstrap Framework

Getting Started with Bootstrap

- There are **two possible ways** to start work with bootstrap
 - Link to a CDN (Content Delivery Networks)
 - Use a Local Copy of Bootstrap

Getting Started with Bootstrap

Link to a CDN

- Bootstrap CDN is a public **content delivery network**. It provides ability to users to load **CSS, JavaScript and images** remotely from its server.
- **Bootstrap CDN Links** to access the scripts and stylesheets used for bootstrap framework utilities. Simply include the code below link in your HTML **head** element and proceed to work with bootstrap.

- **CSS Link**

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css" rel="stylesheet">
```

- **JavaScript Link**

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/js/bootstrap.bundle.min.js"></script>
```

Bootstrap Framework

Link to a CDN: Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css" rel="stylesheet">
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/js/bootstrap.bundle.min.js"></script>
</head>
<body>

<div class="container">
  <h1>My First Bootstrap Page</h1>
  <p>This part is inside a .container class.</p>
  <p>The .container class provides a responsive fixed width container.</p>
</div>

</body>
</html>
```

Note:

Bootstrap uses jQuery for JavaScript plugins (like modals, tooltips, etc). However, if you just use the CSS part of Bootstrap, you don't need jQuery.



Getting Started with Bootstrap

Use a Local Copy of Bootstrap

- Here download your own copy of Bootstrap and integrate it into your project structure.

Steps:

- **Step 1: Download Bootstrap**

- For Download: <https://getbootstrap.com/docs/5.1/getting-started/download/>

- **Step 2: Create a Project**

- Assume project folder (**Bootstrap_Project**) contains two files, **index.html**, **main.css**

- **Step 3: Move Bootstrap Into Your Project Folder**

- Move the newly unzipped **bootstrap-5.1.0-dist** folder into your project folder.

Getting Started with Bootstrap

Use a Local Copy of Bootstrap

Example:

Bootstrap_Project

- | - index.html
- | - main.css
- | - bootstrap-5.1.0-dist
 - | - css
 - | - js

- **Step 4: Link to Your Copy of Bootstrap**

Add below code inside the head tag in index.html

```
<link rel="stylesheet" href="bootstrap-5.1.3-dist/css/bootstrap.min.css"/>  
<script src="bootstrap-5.1.3-dist/js/bootstrap.min.js"></script>
```

Use a Local Copy of Bootstrap: Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet" href="bootstrap-5.1.3-dist/css/bootstrap.min.css"/>
  <script src="bootstrap-5.1.3-dist/js/bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h1>My First Bootstrap Page</h1>
  <p>This part is inside a .container class.</p>
  <p>The .container class provides a responsive fixed width container.</p>
</div>

</body>
</html>
```

Note:

Bootstrap uses jQuery for JavaScript plugins (like modals, tooltips, etc). However, if you just use the CSS part of Bootstrap, you don't need jQuery.



Create First Web Page With Bootstrap: Step by Step

Add HTML5 doctype

- Bootstrap uses **HTML elements** and **CSS properties** that require the **HTML5 doctype**.
- Always include the HTML5 doctype at the beginning of the page, along with the lang attribute and the correct character set:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Bootstrap 5 Example</title>
    <meta charset="utf-8">
  </head>
</html>
```

Create First Web Page With Bootstrap: Step by Step

Responsive meta tag

- Bootstrap is developed ***mobile first***, a strategy in which we **optimize code for mobile devices** first and then scale up components as necessary using CSS media queries.
- To ensure proper rendering and touch zooming for all devices, **add the responsive viewport meta tag** to your **<head>**.

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

- The **width=device-width** part sets the width of the page to follow the **screen-width of the device** (which will vary depending on the device).
- The **initial-scale=1** part sets the **initial zoom level** when the page is first loaded by the browser.

Create First Web Page With Bootstrap: Step by Step

Containers

- Bootstrap also requires a **containing element to wrap site contents**.
- There are **two container classes** to choose from:
- The **.container** class provides a responsive **fixed width container**
- The **.container-fluid** class provides a **full width container**, spanning the entire width of the viewport

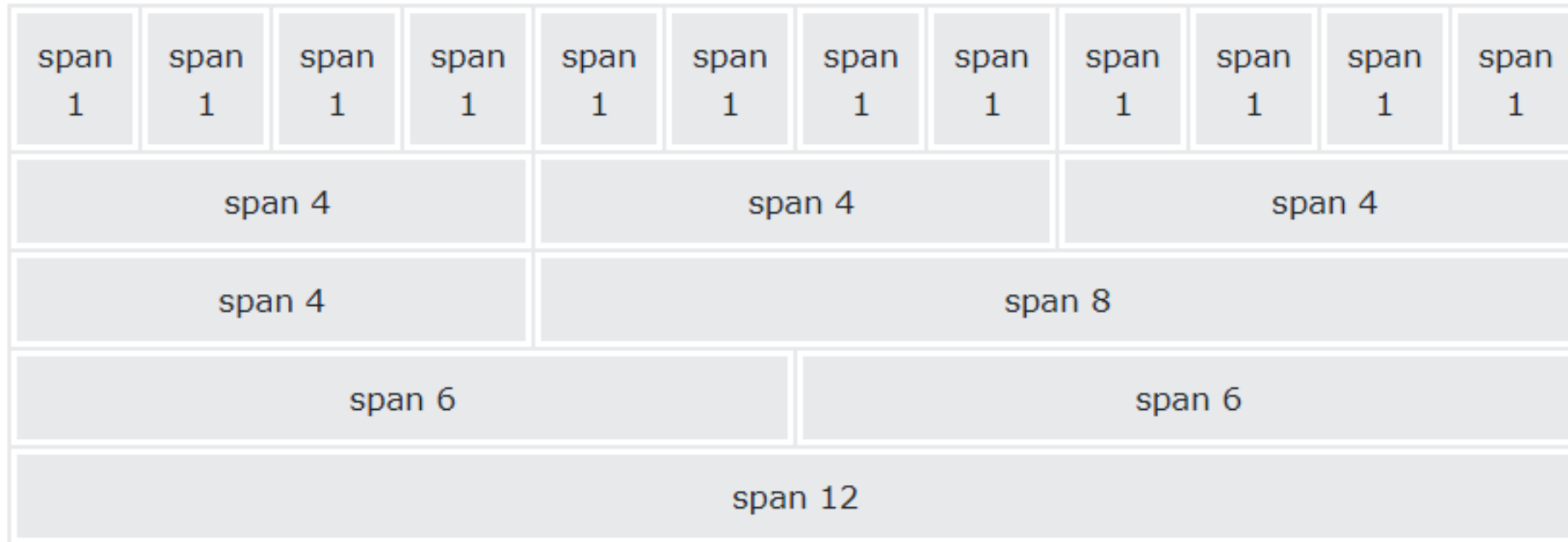


Why Bootstrap over CSS?

- There're no predefined classes for designing in CSS, while Bootstrap has **predefined design classes** that help in quicker and easier designing.
- Bootstrap is useful in **creating responsive designs quickly** as the bootstrap classes take care of responsiveness, while in CSS it takes more effort to have responsiveness using media tags.
- Bootstrap has **more features** to offer for web-design as compared to CSS.

Bootstrap Grids

- Bootstrap's grid system is built with flexbox and allows up to 12 columns across the page.
- You can use all 12 columns **individually** or you can **groups the columns together** to create wider columns.



Bootstrap Grids

The Bootstrap 5 grid system has six classes:

- ✓ **.col-** (extra small devices - screen width less than 576px)
- ✓ **.col-sm-** (small devices - screen width equal to or greater than 576px)
- ✓ **.col-md-** (medium devices - screen width equal to or greater than 768px)
- ✓ **.col-lg-** (large devices - screen width equal to or greater than 992px)
- ✓ **.col-xl-** (xlarge devices - screen width equal to or greater than 1200px)
- ✓ **.col-xxl-** (xxlarge devices - screen width equal to or greater than 1400px)
- These classes **can be combined** to create more dynamic and flexible layouts.

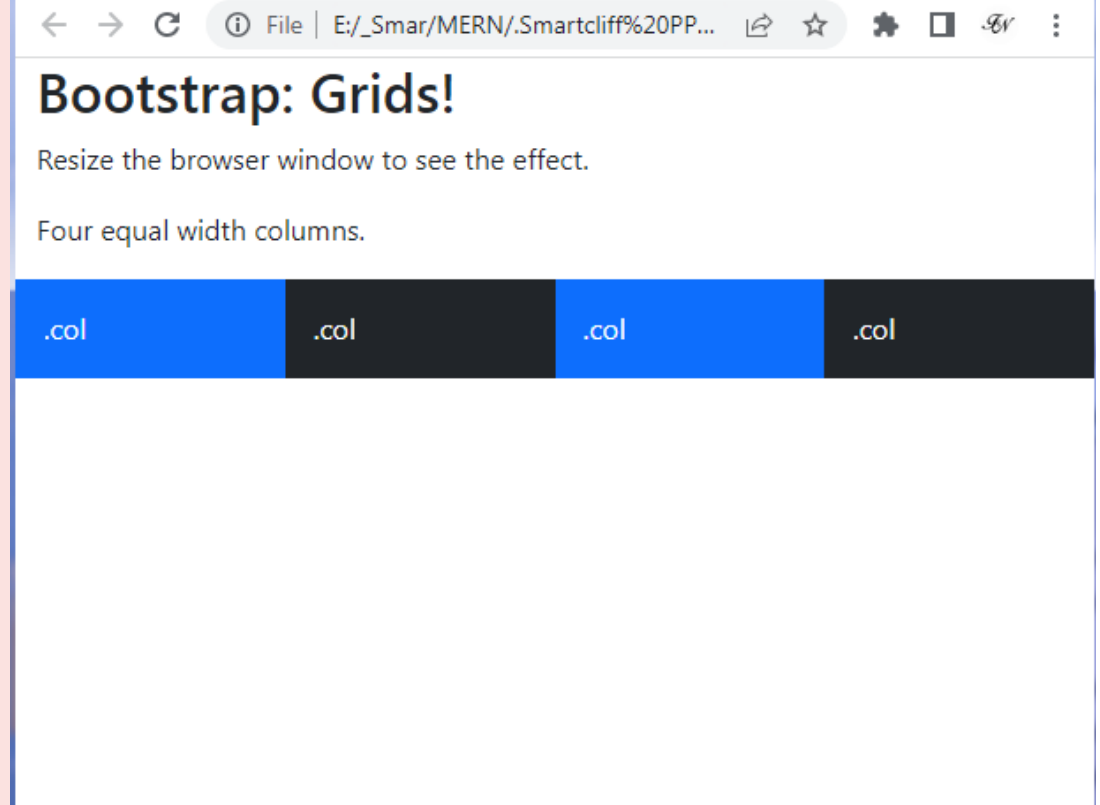
Bootstrap Grids

- First; create a row (**<div class="row">**).
- Then, add the desired number of columns (tags with appropriate **.col-*-*** classes).
- The first star (*) represents the responsiveness: sm, md, lg, xl or xxl, while the second star represents a number, which should add up to 12 for each row.

Bootstrap Grids

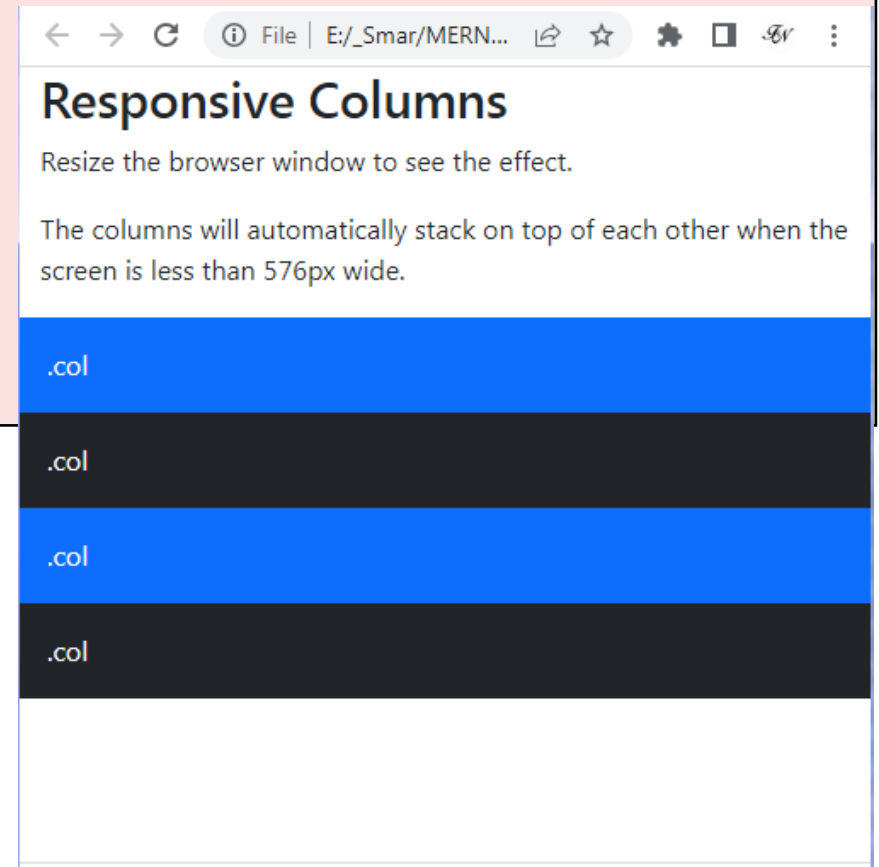
The following example shows how to create four equal-width columns, on all devices and screen widths:

```
<div class="container-fluid">
  <h1>Bootstrap: Grids!</h1>
  <p>Resize the browser window to see the effect.</p>
  <p> Four equal width columns.</p>
  <div class="row">
    <div class="col p-3 bg-primary text-white">.col</div>
    <div class="col p-3 bg-dark text-white">.col</div>
    <div class="col p-3 bg-primary text-white">.col</div>
    <div class="col p-3 bg-dark text-white">.col</div>
  </div>
</div>
```



Bootstrap Grids: Example

```
<div class="container-fluid">
  <h1>Responsive Columns</h1>
  <p>Resize the browser window to see the effect.</p>
  <p>The columns will automatically stack on top of each other when the screen is less than 576px wide.</p>
  <div class="row">
    <div class="col-sm-3 p-3 bg-primary text-white">.col</div>
    <div class="col-sm-3 p-3 bg-dark text-white">.col</div>
    <div class="col-sm-3 p-3 bg-primary text-white">.col</div>
    <div class="col-sm-3 p-3 bg-dark text-white">.col</div>
  </div></div>
```



Bootstrap Typography

- Typography provides some **utilities to add additional styles to texts.**
- **These utilities are:**
 - Text alignment
 - Text transform
 - Font weight and italics
- **Test alignment :**
 - **Align text left:** `<p class="text-start">Start aligned text on all viewport sizes.</p>`
 - **Align text center:** `<p class="text-center">Center aligned text on all viewport sizes.</p>`
 - **Align text right:** `<p class="text-end">End aligned text on all viewport sizes.</p>`

Bootstrap Typography

- **Text Transform:** These classes are used to transform text from one case to another.

- **To transform into lowercase:**

```
<p class="text-lowercase">lowercased text.</p>
```

- **To transform into uppercase:**

```
<p class="text-uppercase">UPPERCASED TEXT.</p>
```

- **To capitalize:**

```
<p class="text-capitalize">Capitalized Text.</p>
```

- **Font size:** These classes are used to change text size. It varies from .fs-1 to .fs-6

```
<p class="fs-1">.fs-1 text</p>  
<p class="fs-2">.fs-2 text</p>  
<p class="fs-3">.fs-3 text</p>  
<p class="fs-4">.fs-4 text</p>  
<p class="fs-5">.fs-5 text</p>  
<p class="fs-6">.fs-6 text</p>
```

Bootstrap Typography

- **Font weight and italics:** To change text in bold or italic form, we use this class.

- **For Bold text:**

```
<p class="fw-bold">Bold text.</p>
```

- **For Bolder weight text:**

```
<p class="fw-bolder">Bolder weight text (relative to the  
parent element).</p>
```

- **For Normal weight text:**

```
<p class="fw-normal">Normal weight text.</p>
```

- **Light weight text**

```
<p class="fw-light">Light weight text.</p>
```

- **Lighter weight text**

```
<p class="fw-lighter">Lighter weight text (relative to the  
parent element).</p>
```

- **Normal weight text**

```
<p class="fsw-normal">Normal weight text.</p>
```

- **Italic text**

```
<p class="fsw-italic">Italic text.</p>
```

Bootstrap Tables

Bootstrap provides following classes to **design tables**:

- `.table`
- `.table-striped`
- `.table-bordered`
- `.table-borderless`
- `.table-dark`
- `.table-hover`
- `.table-responsive`

Bootstrap Tables

Contextual Classes:

To color table rows(<tr>) or table cells(<td>) we can use following classes.

Class	Description
.table-primary	Blue: Indicates an important action
.table-success	Green: Indicates a successful or positive action
.table-danger	Red: Indicates a dangerous or potentially negative action
.table-info	Light blue: Indicates a neutral informative change or action
.table-warning	Orange: Indicates a warning that might need attention
.table-active	Grey: Applies the hover color to the table row or table cell
.table-secondary	Grey: Indicates a slightly less important action
.table-light	Light grey table or table row background
.table-dark	Dark grey table or table row background

Bootstrap Tables: Example

```
<table class="table">
  <thead>
    <tr>
      <th>Firstname</th><th>Lastname</th><th>Email</th>
    </tr>
  </thead> <tbody>
    <tr>
      <td>Default</td><td>Defaultson</td>
      <td>def@somemail.com</td>
    </tr>
    <tr class="table-primary">
      <td>Primary</td><td>Joe</td>
      <td>joe@example.com</td>
    </tr>
    <tr class="table-success">
      <td>Success</td> <td>Doe</td>
      <td>john@example.com</td>
    </tr>
    <tr class="table-danger">
      <td>Danger</td><td>Moe</td>
      <td>mary@example.com</td>
    </tr>
```

```
<tr class="table-info">
  <td>Info</td><td>Dooley</td>
  <td>july@example.com</td>
</tr>
<tr class="table-warning">
  <td>Warning</td><td>Refs</td>
  <td>bo@example.com</td>
</tr>
<tr class="table-active">
  <td>Active</td><td>Activeson</td>
  <td>act@example.com</td>
</tr>
<tr class="table-secondary">
  <td>Secondary</td><td>Secondson</td>
  <td>sec@example.com</td>
</tr>
<tr class="table-light">
  <td>Light</td> <td>Angie</td>
  <td>angie@example.com</td>
</tr>
<tr class="table-dark">
  <td>Dark</td><td>Bo</td>
  <td>bo@example.com</td>
</tr> </tbody></table>
```

Bootstrap Tables: Example

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Contextual Classes

Contextual classes can be used to color the table, table rows or table cells.

Firstname	Lastname	Email
Default	Defaultson	def@somemail.com
Primary	Joe	joe@example.com
Success	Doe	john@example.com
Danger	Moe	mary@example.com
Info	Dooley	july@example.com
Warning	Refs	bo@example.com
Active	Activeson	act@example.com
Secondary	Secondson	sec@example.com
Light	Angie	angie@example.com
Dark	Bo	bo@example.com

Bootstrap Images

Bootstrap Image Shapes: To define shape of an image we use following classes.

- **.rounded**

```

```

- **.rounded-circle**

```

```

- **.img-thumbnail**

```

```

Aligning Images :

- Float an image to the left with the `.float-start` class or to the right with `.float-end`:

```
  

```

Bootstrap Images

Centered Image:

- Center an image by adding the utility classes `.mx-auto` (margin:auto) and `.d-block` (display:block) to the image:

```

```

Responsive Image :

- We use responsive image when we want images to **fit according to screen size**. We can create responsive image by adding **.img-fluid** class to the `<img>` tag.

Image Gallery:

```
img class="img-fluid" src="img_fluid.jpg" alt="Responsive_img">
```

- One way to make image gallery is to use **Bootstrap's grid system** with **.thumbnail** class.

Bootstrap Alerts

- Bootstrap alerts are used to **show predefined alert message**.
- This looks stylish and attracts more attention .
- Alerts are created with the **.alert** class, followed by one of the contextual classes
 - .alert-success
 - .alert-info
 - .alert-warning
 - .alert-danger
 - .alert-primary
 - .alert-secondary
 - .alert-light
 - .alert-dark

Bootstrap Alerts

```
<div class="alert alert-success">
  <strong>Success!</strong> This alert box indicates a successful or positive action.
</div>
<div class="alert alert-info">
  <strong>Info!</strong> This alert box indicates a neutral informative change or action.
</div>
<div class="alert alert-warning">
  <strong>Warning!</strong> This alert box indicates a warning that might need attention.
</div>
<div class="alert alert-danger">
  <strong>Danger!</strong> This alert box indicates a dangerous or potentially negative action.
</div>
```

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Success! This alert box indicates a successful or positive action.

Info! This alert box indicates a neutral informative change or action.

Warning! This alert box indicates a warning that might need attention.

Danger! This alert box indicates a dangerous or potentially negative action.

Bootstrap Alerts

```
<div class="alert alert-primary">
  <strong>Primary!</strong> This alert box indicates an important action.
</div>
<div class="alert alert-secondary">
  <strong>Secondary!</strong> This alert box indicates slightly less important action.
</div>
<div class="alert alert-dark">
  <strong>Dark!</strong> This alert box indicates dark grey alert.
</div>
<div class="alert alert-light">
  <strong>Light</strong> This alert box indicates light grey alert.
</div>
```

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Primary! This alert box indicates an important action.

Secondary! This alert box indicates slightly less important action.

Dark! This alert box indicates dark grey alert.

Light This alert box indicates light grey alert.

Bootstrap Buttons

Button Style : In Bootstrap to create different types styles of button we use following classes.

- .btn
- .btn-primary
- .btn-secondary
- .btn-success
- .btn-info
- .btn-warning
- .btn-danger
- .btn-dark
- .btn-light
- .btn-link

```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-link">Link</button>
```



Bootstrap Buttons

Button Outline : In Bootstrap to create different types outline/bordered buttons we use following classes.

Move the mouse over them to see an additional "hover" effect:

- .btn
- .btn-outline-primary
- .btn-outline-secondary
- .btn-outline-success
- .btn-outline-info
- .btn-outline-warning
- .btn-outline-danger
- .btn-outline-dark
- .btn-outline-light

```
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-secondary">Secondary</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
<button type="button" class="btn btn-outline-light text-dark">Light</button>
```

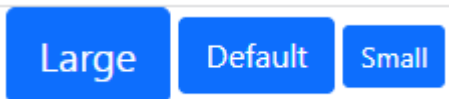


Bootstrap Buttons

Button Size : To define button size we use following classes.

- .btn-lg class for large buttons
- .btn-sm class for small buttons

```
<button type="button" class="btn btn-primary btn-lg">Large</button>  
<button type="button" class="btn btn-primary">Default</button>  
<button type="button" class="btn btn-primary btn-sm">Small</button>
```



Bootstrap Buttons

Block Level Buttons : This type of button has width equal to its parent element.

- .d-grid

```
<div class="d-grid">  
  <button type="button" class="btn btn-primary btn-block">Full-Width Button</button>  
</div>
```

Full-Width Button

Active/Disabled Buttons : To make such buttons we use following classes.

- .active

- .disabled

```
<button type="button" class="btn btn-primary active">Active Primary</button>  
<button type="button" class="btn btn-primary" disabled>Disabled Primary</button>  
<a href="#" class="btn btn-primary disabled">Disabled Link</a>
```

Active Primary

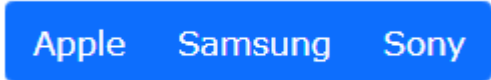
Disabled Primary

Bootstrap Buttons

Button Groups : Using this feature we can create group of buttons.

- .btn-group

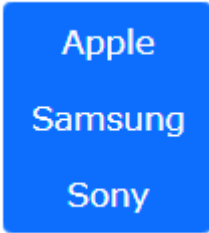
```
<div class="btn-group">
  <button type="button" class="btn btn-primary">Apple</button>
  <button type="button" class="btn btn-primary">Samsung</button>
  <button type="button" class="btn btn-primary">Sony</button>
</div>
```


 A horizontal group of three blue buttons with white text. The buttons are labeled "Apple", "Samsung", and "Sony" from left to right.

Verticle Button Group

- .btn-group-vertical

```
<div class="btn-group-vertical">
  <button type="button" class="btn btn-primary">Apple</button>
  <button type="button" class="btn btn-primary">Samsung</button>
  <button type="button" class="btn btn-primary">Sony</button>
</div>
```


 A vertical group of three blue buttons with white text. The buttons are labeled "Apple", "Samsung", and "Sony" from top to bottom.

Bootstrap Badges

Badges: Badges are used to add additional information to any content.

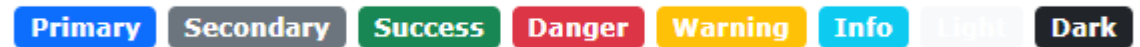
- **.badge** class together with a contextual class (like **.bg-secondary**) within **** elements to create rectangular badges.

```
<h1>Example heading <span class="badge bg-secondary">New</span></h1>
<h6>Example heading <span class="badge bg-secondary">New</span></h6>
```



Contextual Badges: Use any of the contextual classes (**.bg-***) to change the color of a badge.

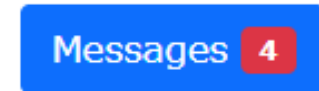
```
<span class="badge bg-primary">Primary</span>
<span class="badge bg-secondary">Secondary</span>
<span class="badge bg-success">Success</span>
<span class="badge bg-danger">Danger</span>
<span class="badge bg-warning">Warning</span>
<span class="badge bg-info">Info</span>
<span class="badge bg-light">Light</span>
<span class="badge bg-dark">Dark</span>
```



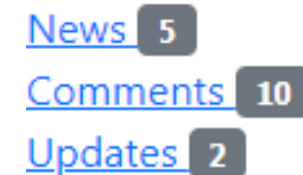
Bootstrap Badges

Badge inside an element :

```
<button type="button" class="btn btn-primary">
  Messages <span class="badge bg-danger">4</span>
</button>
```



```
<div>
<a href="#">News <span class="badge bg-secondary">5</span></a><br>
<a href="#">Comments <span class="badge bg-secondary">10</span></a><br>
<a href="#">Updates <span class="badge bg-secondary">2</span></a>
</div>
```



Bootstrap Progress Bar

Progress Bar : A progress bar can be used to show how far a user is in a process.

- Use **Class .progress** within a **<div>**.
- Add the **.progress-bar** class to its child element.
- Use the CSS **width** property to set the width of the progress bar .

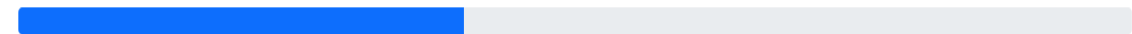
Default Progress Bar:

```
<div class="progress">  
  <div class="progress-bar" style="width:70%"></div>  
</div>
```



Progress Bar Height: Use the CSS **height** property to change it.

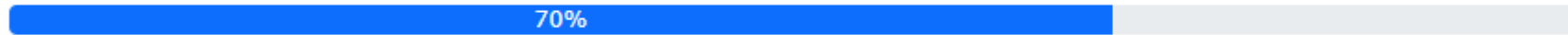
```
<div class="progress" style="height:20px">  
  <div class="progress-  
bar" style="width:40%;height:20px"></div>  
</div>
```



Bootstrap Progress Bar

Progress bar Labels: Add text inside the progress bar to show the visible percentage.

```
<div class="progress">
  <div class="progress-bar" style="width:70%">70%</div>
</div>
```



Colored Progress Bars: By default, the progress bar is blue (primary). Use any of the contextual background classes to change its color.

```
<div class="progress">
  <div class="progress-bar" style="width:70%">70%</div>
</div>
<!-- Red -->
<div class="progress">
  <div class="progress-bar bg-danger" style="width:50%"></div>
</div>
```

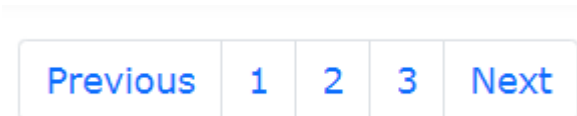


Bootstrap Pagination

Bootstrap Pagination : To sort the web pages in an organized manner we use **Bootstrap Pagination**.

- **Pagination**: Add this class to get the pagination on your website.
- To create a basic pagination, add the **.pagination** class to an **** element.
- Then add the **.page-item** to each **** element and a **.page-link** class to each link inside ****

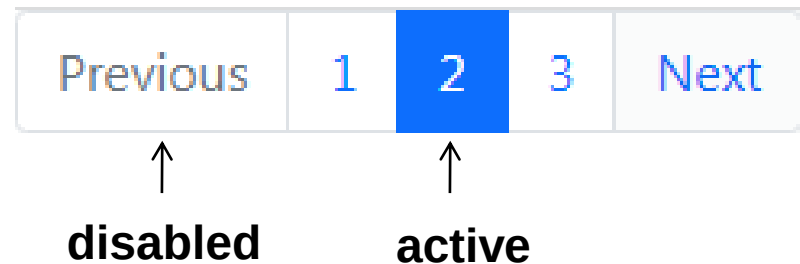
```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



Bootstrap Pagination

- **Active State:** The **.active** class is used to "highlight" the current page.
- **Disabled State:** The **.disabled** class is used for un-clickable links.
 - Use with ****.

```
<ul class="pagination">
  <li class="page-item disabled"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item active"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```



Bootstrap Pagination

Pagination size: Pagination blocks can also be sized to a larger or a smaller size

- **.pagination-lg** for larger blocks
- **.pagination-sm** for smaller blocks
 - Use with ****.

```
<ul class="pagination pagination-lg">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
<ul class="pagination pagination-sm">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li></ul>
```



Bootstrap List Groups

List Group: To create group of list with list items we use **Bootstrap list group**.

- Use **class “.list-group”** within the **** element.
- Use **class “.list-group-item”** within the **** element

```
<ul class="list-group">  
  <li class="list-group-item">First item</li>  
  <li class="list-group-item">Second item</li>  
  <li class="list-group-item">Third item</li>  
</ul>
```

First item

Second item

Third item

Bootstrap List Groups

Bootstrap List group with Badges: To add badges in a list group we follow following procedure:

- Create a **** element with **class."badge"**.

```
<ul class="list-group">
  <li class="list-group-item d-flex justify-content-between align-items-center">
    Inbox
    <span class="badge bg-primary rounded-pill">12</span>
  </li>
  <li class="list-group-item d-flex justify-content-between align-items-center">
    Ads
    <span class="badge bg-primary rounded-pill">50</span>
  </li>
  <li class="list-group-item d-flex justify-content-between align-items-center">
    Junk
    <span class="badge bg-primary rounded-pill">99</span>
  </li>
</ul>
```

Inbox	12
Ads	50
Junk	99

Bootstrap Dropdown

Bootstrap Dropdown: To make **toggleable menus** we use **Bootstrap dropdown class**.

- The **.dropdown** class indicates a dropdown menu.
- To open the dropdown menu, use a button or a link with a class of **.dropdown-toggle** and the **data-bs-toggle="dropdown"** attribute.
- Add the **.dropdown-menu** class to a **<div>** element to actually build the dropdown menu.
- Then add the **.dropdown-item** class to each element (links or buttons) inside the dropdown menu.

```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-
toggle" data-bs-toggle="dropdown">
    Dropdown Example
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">HTML</a></li>
    <li><a class="dropdown-item" href="#">JavaScript</a></li>
  </ul>
</div>
```



Normal



On Click

Bootstrap Dropdown

Bootstrap Dropdown Divider: The `.dropdown-divider` class is used to separate links inside the dropdown menu with a thin horizontal border:

```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-toggle" data-bs-
toggle="dropdown">
    Dropdown Example
  </button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">HTML</a></li>
    <li><a class="dropdown-item" href="#">JavaScript</a></li>
    <li><hr class="dropdown-divider"></li>
    <li><a href="#">About Us</a></li>
  </ul>
</div>
```



Bootstrap Collapse

- Collapsibles are useful when you want to hide and show large amount of content.
- Use **“.collapse”** class

```
<button data-bs-toggle="collapse" data-bs-target="#demo"> Simple collapsible </button>
```

```
<div id="demo" class="collapse">
```

Content you want to hide. Mostly we hide large content if It is not that much important for all users or sometime we hide content to fit more content in small webpage.

```
</div>
```

Normal View

Simple collapsible

On Click

Simple collapsible

Content you want to hide. Mostly we hide large content if It is not that much important for all users or sometime we hide content to fit more content in small webpage.

Bootstrap Collapse

- By default, the collapsible content is hidden.
- Add **.show** class to show the content by default:

```
<button data-bs-toggle="collapse" data-bs-target="#demo"> Simple collapsible </button>

<div id="demo" class="collapse show " >
    Content you want to hide. Mostly we hide large content if It is not that much important for all users
    or sometime we hide content to fit more content in small webpage.
</div>
```

Normal View

Simple collapsible

Content you want to hide. Mostly we hide large content if It is not that much important for all users or sometime we hide content to fit more content in small webpage.

On Click

Simple collapsible

Bootstrap Framework

Bootstrap Flex

Bootstrap Flexbox: Bootstrap 5 uses flexbox, instead of floats, to handle the layout.

To create a flexbox container and to transform direct children into flex items, use the d-flex class:

```
<div class="d-flex p-3 bg-secondary text-white">
  <div class="p-2 bg-info">Flex item 1</div>
  <div class="p-2 bg-warning">Flex item 2</div>
  <div class="p-2 bg-primary">Flex item 3</div>
</div>
```



Use **.flex-column** to display the flex items vertically (on top of each other)

```
<div class="d-flex flex-column">
  <div class="p-2 bg-info">Flex item 1</div>
  <div class="p-2 bg-warning">Flex item 2</div>
  <div class="p-2 bg-primary">Flex item 3</div>
</div>
```



Bootstrap Forms: Stacked Form

- All textual `<input>` and `<textarea>` elements with class **.form-control** get proper form styling.
- Add a **.form-label** class to each label element to ensure correct padding.
- Checkboxes have different markup. They are wrapped around a container element with **.form-check**, and labels have a class of **.form-check-label**, while checkboxes and radio buttons use **.form-check-input**.

Bootstrap Forms: Stacked Form

```
<div class="container mt-3">
  <h2>Stacked form</h2>
  <form>
    <div class="mb-3 mt-3">
      <label for="email">Email:</label>
      <input type="email" class="form-control" id="email" placeholder="Enter email" name="email">
    </div>
    <div class="mb-3">
      <label for="pwd">Password:</label>
      <input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pswd">
    </div>
    <div class="form-check mb-3">
      <label class="form-check-label">
        <input class="form-check-input" type="checkbox" name="remember"> Remember me
      </label>
    </div>
    <button type="submit" class="btn btn-primary">Submit</button>
  </form>
</div>
```


Bootstrap Forms: Stacked Form

← → ↻ ⓘ File | E:/_Smar/MERN/.Smartcliff%20PPT/Bootstrap/16.Forms.h... 🔗 ☆ ⚙️ □ Ⓔ ⋮

Stacked form

Email:

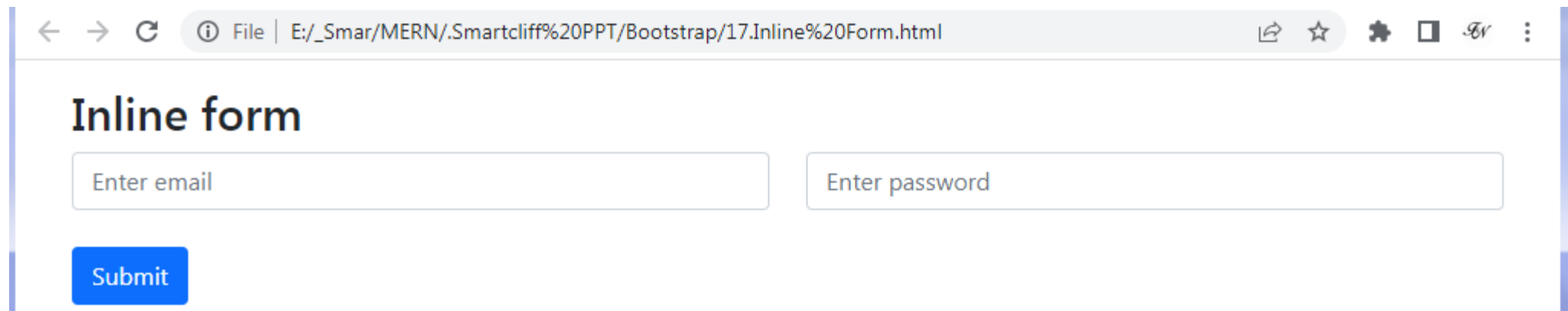
Password:

☐ Remember me

Bootstrap Forms: Row/Grid (Inline Forms)

To display the form elements to appear side by side, use **.row** and **.col**

```
<form>
  <div class="row">
    <div class="col">
      <input type="text" class="form-control" placeholder="Enter email" name="email">
    </div>
    <div class="col">
      <input type="password" class="form-control" placeholder="Enter password" name="pswd">
    </div>
  </div> <button type="submit" class="btn btn-primary">Submit</button>
</form>
```



The screenshot shows a web browser window with the address bar displaying the file path: E:/_Smar/MERN/.Smartcliff%20PPT/Bootstrap/17.Inline%20Form.html. The page title is "Inline form". The form consists of two input fields side-by-side. The first field is labeled "Enter email" and the second field is labeled "Enter password". Below these fields is a blue "Submit" button.

Bootstrap Forms: Textarea

- Use the **.form-control** class to style textareas

```
<div class="container mt-3">  
<form >  
  <div class="mb-3 mt-3">  
    <label for="comment">Comments:</label>  
    <textarea class="form-control" rows="5" id="comment" name="text"></textarea>  
  </div>  
  <button type="submit" class="btn btn-primary">Submit</button>  
</form>  
</div>
```

Comments:

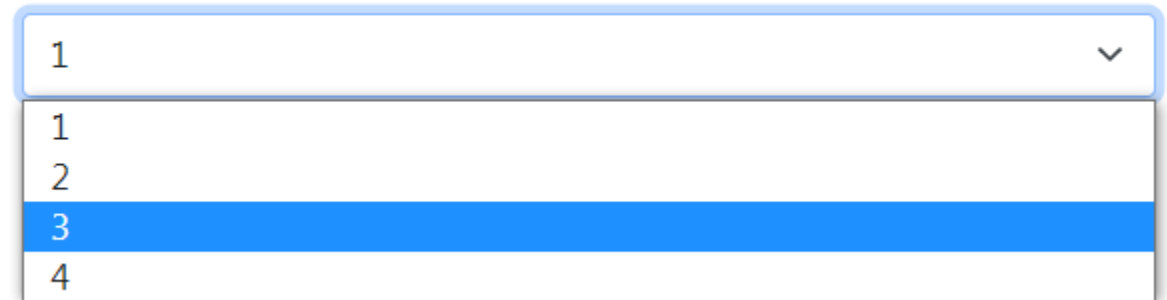
Submit

Bootstrap Forms: Select Menu

- Select menus are used to allow the user to pick from multiple options.
- To style a select menu in Bootstrap 5, add the **.form-select** class to the `<select>` element

```
<div class="container mt-3">
<form >
  <label for="sel1" class="form-label">Select list (select one):</label>
  <select class="form-select" id="sel1" name="sellist1">
    <option>1</option>
    <option>2</option>
    <option>3</option>
    <option>4</option>
  </select>
</form>
</div>
```

Select list (select one):



Bootstrap Forms: Multiple-Select Menu

- To style a select menu in Bootstrap 5, add the **.form-select** class to the <select> element.
- Multiple select menu (hold ctrl or shift (or drag with the mouse) to select more than one).

```
<div class="container mt-3">
<form >
  <label for="sel2" class="form-label">Mutiple select list (hold shift to select
more than one):</label>
  <select multiple class="form-select" id="sel2" name="sellist2">
    <option>1</option>
    <option>2</option>
    <option>3</option>
    <option>4</option>
    <option>5</option>
  </select>
</form>
</div>
```

Mutiple select list (hold shift to select more than one):



Bootstrap Forms: Checkboxes

- Checkboxes are used if you want the user to select any number of options from a **list of preset options**.

```
<div class="container mt-3">
  <h2>Checkboxes</h2>
  <form action="/action_page.php">
    <div class="form-check">
      <input type="checkbox" class="form-check-input" id="check1" name="option1" value="something"
checked>
      <label class="form-check-label" for="check1">Option 1</label>
    </div>
    <div class="form-check">
      <input type="checkbox" class="form-check-input" id="check2" name="option2" value="something">
      <label class="form-check-label" for="check2">Option 2</label>
    </div>
    <div class="form-check">
      <input type="checkbox" class="form-check-input" disabled>
      <label class="form-check-label">Option 3</label>
    </div>
    <button type="submit" class="btn btn-primary mt-3">Submit</button>
  </form>
</div>
```

Checkboxes

- ☒ Option 1
- ☐ Option 2
- ☐ Option 3

Submit

Bootstrap Forms: Radio Buttons

- Radio buttons are used to **limit the user** at just one selection from a list of preset options.

```
<div class="container mt-3">
  <h2>Radio buttons</h2>
  <form>
    <div class="form-check">
      <input type="radio" class="form-check-input" id="radio1" name="optradio" value="option1" checked>
      <label class="form-check-label" for="radio1">Option 1</label>
    </div>
    <div class="form-check">
      <input type="radio" class="form-check-input" id="radio2" name="optradio" value="option2">
      <label class="form-check-label" for="radio2">Option 2</label>
    </div>
    <div class="form-check">
      <input type="radio" class="form-check-input" disabled>
      <label class="form-check-label">Option 3</label>
    </div>
    <button type="submit" class="btn btn-primary mt-3">Submit</button>
  </form>
</div>
```

Radio buttons

- ☒ Option 1
☐ Option 2
☐ Option 3

Submit

Bootstrap Forms: File input

```
<div class="mb-3">
  <label for="formFile" class="form-label">Default file input example</label>
  <input class="form-control" type="file" id="formFile">
</div>
<div class="mb-3">
  <label for="formFileMultiple" class="form-label">Multiple files input example</label>
  <input class="form-control" type="file" id="formFileMultiple" multiple>
</div>
<div class="mb-3">
  <label for="formFileDisabled" class="form-label">Disabled file input example</label>
  <input class="form-control" type="file" id="formFileDisabled" disabled>
</div>
```

Default file input example

Choose File

No file chosen

Multiple files input example

Choose Files

No file chosen

Disabled file input example

Choose File

No file chosen

Bootstrap Forms: Form Validation

There are different validation classes to provide valuable feedback to users.

- Add either **.was-validated** or **.needs-validation** to the `<form>` element, depending on whether to provide validation feedback before or after submitting the form.
- The input fields will have a green (valid) or red (invalid) border to indicate what's missing in the form.
- Add a **.valid-feedback** or **.invalid-feedback** message to tell the user explicitly what's missing, or needs to be done before submitting the form.

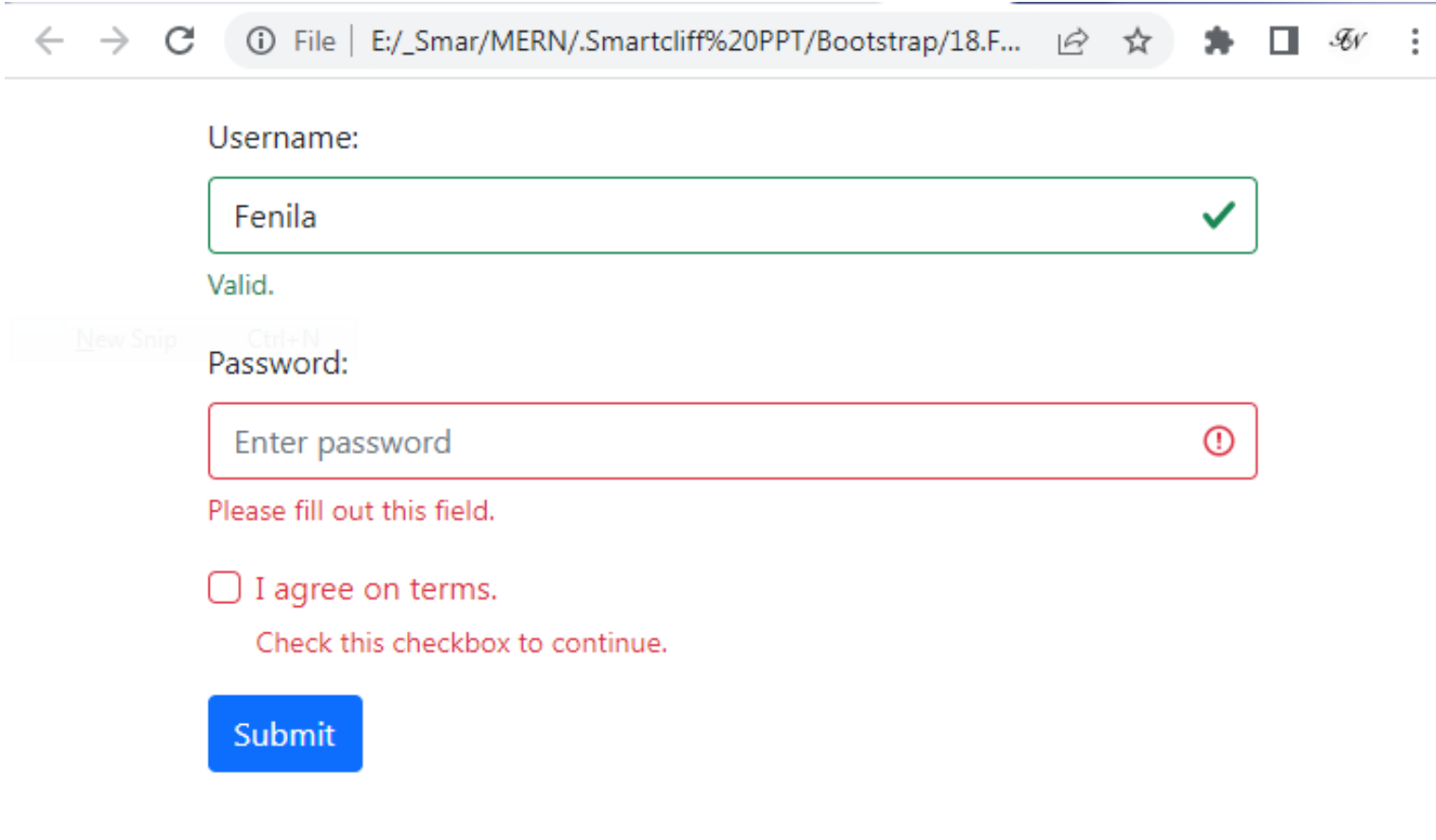
Bootstrap Forms: Form Validation

```
<form class="was-validated">
  <div class="mb-3 mt-3">
    <label for="uname" class="form-label">Username:</label>
    <input type="text" class="form-control" id="uname" placeholder="Enter username" name="uname" required>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>

  <div class="mb-3">
    <label for="pwd" class="form-label">Password:</label>
    <input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pswd" required>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>

  <div class="form-check mb-3">
    <input class="form-check-input" type="checkbox" id="myCheck" name="remember" required>
    <label class="form-check-label" for="myCheck">I agree on terms.</label>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Check this checkbox to continue.</div>
  </div> <button type="submit" class="btn btn-primary">Submit</button></form>
```

Bootstrap Forms: Form Validation



The screenshot shows a web browser window with a Bootstrap form. The browser's address bar displays the file path: E:/_Smar/MERN/.Smartcliff%20PPT/Bootstrap/18.F... The form contains two input fields. The first field, labeled 'Username:', contains the text 'Fenila' and has a green checkmark icon on the right, indicating it is valid. Below this field, the text 'Valid.' is displayed in green. The second field, labeled 'Password:', contains the placeholder text 'Enter password' and has a red exclamation mark icon on the right, indicating it is invalid. Below this field, the text 'Please fill out this field.' is displayed in red. Below the password field, there is a checkbox with the text 'I agree on terms.' and a red message 'Check this checkbox to continue.' Below the checkbox is a blue 'Submit' button. On the left side of the browser window, a 'New Snip' button with a 'Ctrl+N' shortcut is visible.

Username:

Fenila ✓

Valid.

New Snip Ctrl+N

Password:

Enter password !

Please fill out this field.

☐ I agree on terms.
Check this checkbox to continue.

Submit

Bootstrap Navs

- To create a simple horizontal menu, add the **.nav** class to a **** element, followed by **.nav-item** for each **** and add the **.nav-link** class to their links

```
<h3>WebsiteName</h3>
<ul class="nav">
  <li class="nav-item">
    <a class="nav-link" href="#">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page1</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page2</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

WebsiteName

[Home](#) [Page1](#) [Page2](#) Disabled

Bootstrap Navs

- Turn the nav menu into navigation tabs with the **.nav-tabs** class. Add the **.active** class to the active/current link.

```
<h3>WebsiteName</h3>
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a class="nav-link active" href="#">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page1</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page2</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

WebsiteName

[Home](#)[Page1](#)[Page2](#)[Disabled](#)

Bootstrap Navs

- Add the **.flex-column** class to create a vertical nav.

```
<h3>WebsiteName</h3>
<ul class="nav flex-column">
  <li class="nav-item">
    <a class="nav-link active" href="#">Home</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page1</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Page2</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

WebsiteName

[Home](#)[Page1](#)[Page2](#)[Disabled](#)

Bootstrap Navbar

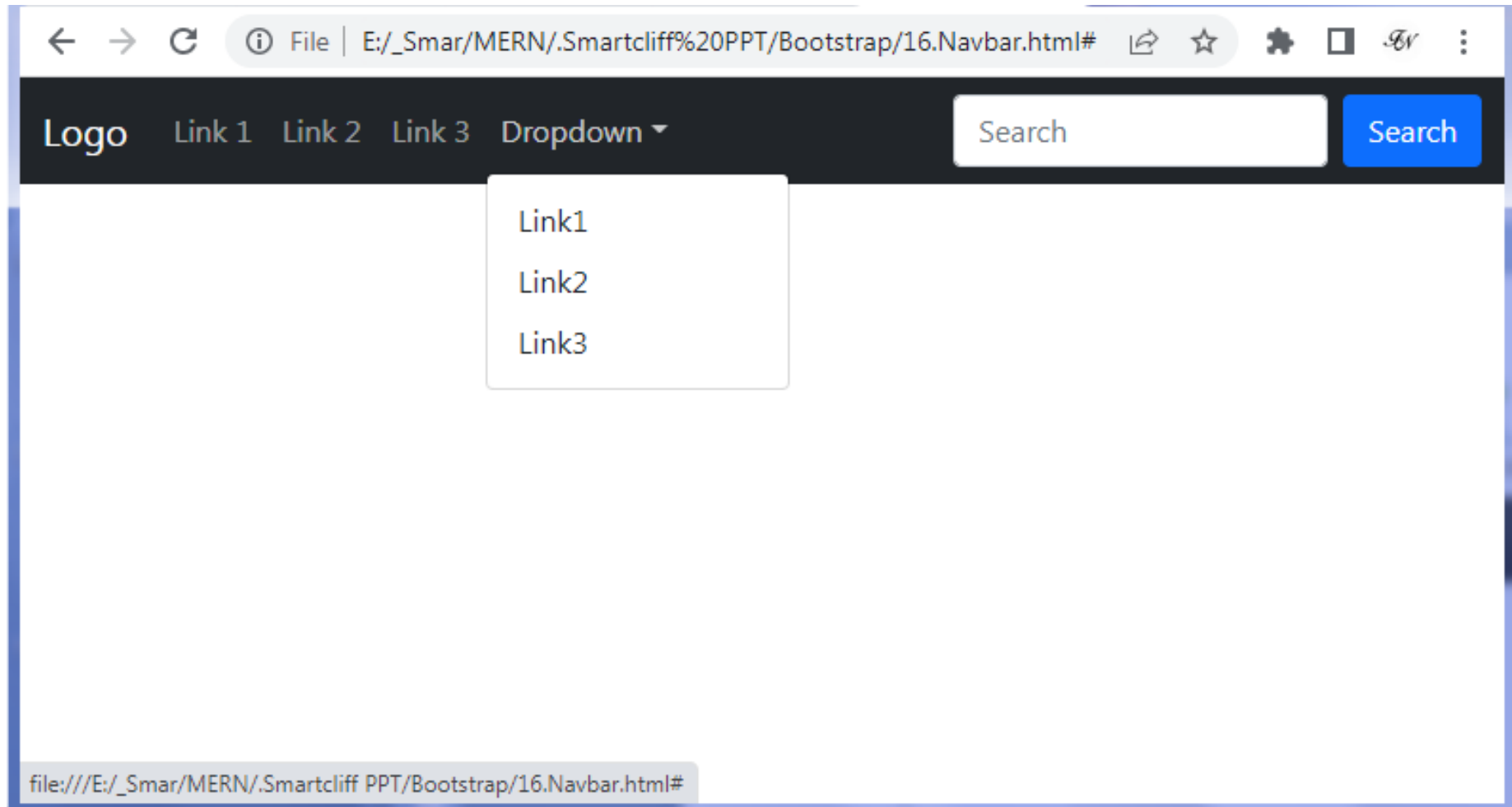
- A navigation bar is a navigation header that is placed at the top of the page.
- A standard navigation bar is created with the **.navbar** class, followed by a responsive collapsing class: **.navbar-expand-xxl|xl|lg|md|sm** (stacks the navbar vertically on xxlarge, extra large, large, medium or small screens).
- To add links inside the navbar, use either an **** element (or a **<div>**) with **class="navbar-nav"**.
- Add **** elements with a **.nav-item** class followed by an **<a>** element with a **.nav-link** class.
- The **.navbar-brand** class is used to highlight the brand/logo/project name of our page.

Bootstrap Navbar

```
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Logo</a>
    <div class="collapse navbar-collapse" id="collapsibleNavbar">
      <ul class="navbar-nav me-auto">
        <li class="nav-item">
          <a class="nav-link" href="#">Link 1</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Link 2</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Link 3</a>
        </li>
        <li class="nav-item dropdown">
          <a class="nav-link dropdown-toggle"
href="#" role="button" data-bs-toggle="dropdown">Dropdown</a>
          <ul class="dropdown-menu">
            <li><a class="dropdown-item" href="#">Link1</a></li>
            <li><a class="dropdown-item" href="#">Link2</a></li>
```

```
            </li>
            </ul>
          </li>
        </ul>
      </div>
    </div>
    <form class="d-flex">
      <input class="form-control me-2" type="text"
placeholder="Search">
      <button class="btn btn-primary"
type="button">Search</button>
    </form>
  </div>
</nav>
```


Bootstrap Navbar



Bootstrap Modals

The Modal component is a dialog box/popup window that is displayed on top of the current page.

```
<div class="container mt-3">
  <h3>Modal Example</h3>
  <p>Click on the button to open the modal.</p>

  <!-- Button to Open the Modal -->
  <button type="button" class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
    Open modal
  </button>
</div>

<!-- The Modal -->
<div class="modal" id="myModal">
  <div class="modal-dialog">
    <div class="modal-content">
      <!-- Modal Header -->
      <div class="modal-header">
        <h4 class="modal-title">Modal Heading</h4>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
```

Bootstrap Modals

```
<!-- Modal body -->
<div class="modal-body">
  Modal body..
</div>

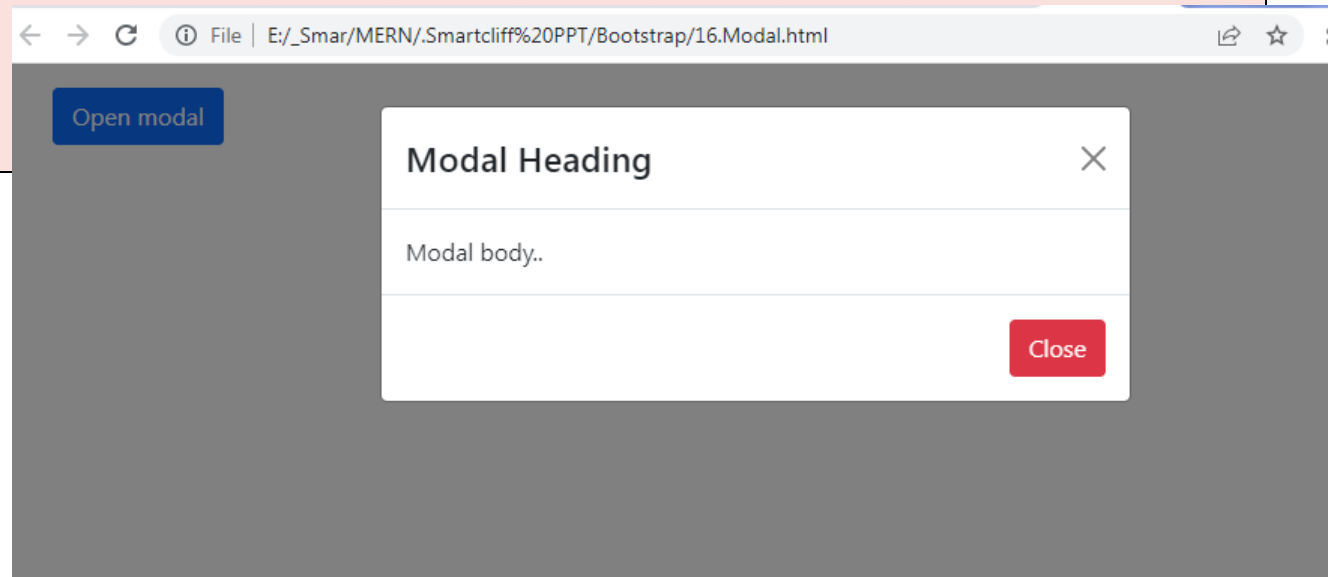
<!-- Modal footer -->
<div class="modal-footer">
  <button type="button" class="btn btn-danger" data-bs-dismiss="modal">Close</button>
</div>

</div>
</div>
</div>
```

Open modal

Normal

On Click ➡



Bootstrap Modals

Bootstrap Modal size : .

- Change the size of the modal by adding,
 - **.modal-sm** class for small modals (max-width 300px),
 - **.modal-lg** class for large modals (max-width 800px),
 - **.modal-xl** for extra large modals (max-width 1140px).
 - Default is 500px max-width
- Add size classes to the **<div>** element with class **.modal-dialog**
- The modal to span the whole width and height of the page, use the **.modal-fullscreen** along with **.modal-dialog** class

Bootstrap Carousel / Slideshow

Bootstrap Carousel : To add Slider to your webpage you can use **Bootstrap Carousel**.

Class	Description
.carousel	Creates a carousel
.carousel-indicators	Adds indicators for the carousel. These are the little dots at the bottom of each slide
.carousel-inner	Adds slides to the carousel
.carousel-item	Specifies the content of each slide
.carousel-control-prev	Adds a left (previous) button to the carousel, which allows the user to go back between the slides
.carousel-control-next	Adds a right (next) button to the carousel, which allows the user to go forward between the slides
.carousel-control-prev-icon	Used together with .carousel-control-prev to create a "previous" button
.carousel-control-next-icon	Used together with .carousel-control-next to create a "next" button
.slide	Adds a CSS transition and animation effect when sliding from one item to the next. Remove this class if you do not want this effect

Bootstrap Carousel / Slideshow

```

<!-- Carousel -->
<div id="demo" class="carousel slide" data-bs-
ride="carousel">

<!-- Indicators/dots -->
<div class="carousel-indicators">
  <button type="button" data-bs-target="#demo" data-bs-
slide-to="0" class="active"></button>
  <button type="button" data-bs-target="#demo" data-bs-
slide-to="1"></button>
  <button type="button" data-bs-target="#demo" data-bs-
slide-to="2"></button>
</div>

<!-- The slideshow/carousel -->
<div class="carousel-inner">
  <div class="carousel-item active">
    
  </div>
  <div class="carousel-item">
    
  </div>
  <div class="carousel-item">
    
  </div>
</div>

<!-- Left and right controls/icons -->
<button class="carousel-control-
prev" type="button" data-bs-target="#demo" data-bs-
slide="prev">
  <span class="carousel-control-prev-icon"></span>
</button>
<button class="carousel-control-
next" type="button" data-bs-target="#demo" data-bs-
slide="next">
  <span class="carousel-control-next-icon"></span>
</button>
</div>

```

```

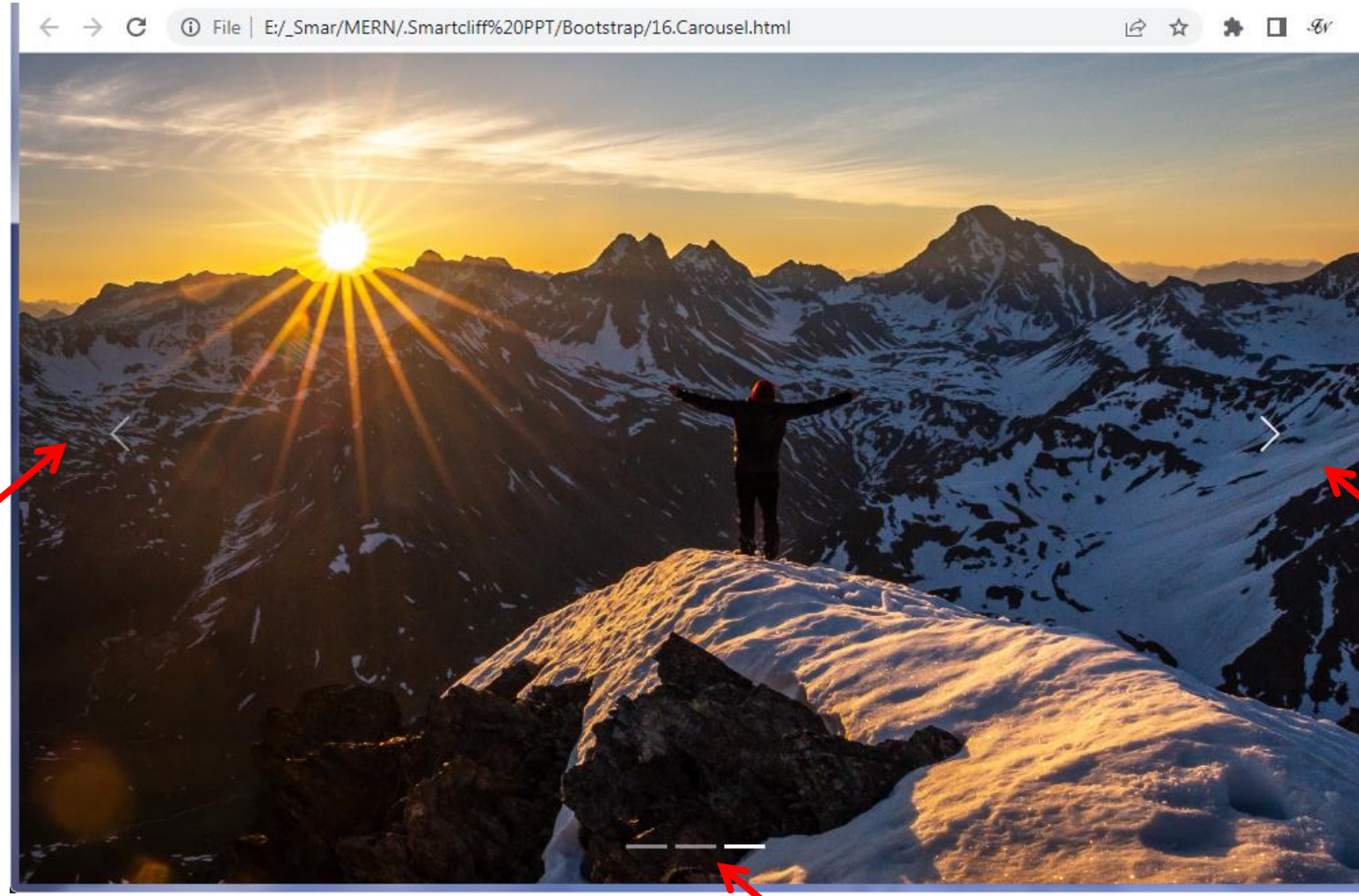
    
  </div>

<!-- Left and right controls/icons -->
<button class="carousel-control-
prev" type="button" data-bs-target="#demo" data-bs-
slide="prev">
  <span class="carousel-control-prev-icon"></span>
</button>
<button class="carousel-control-
next" type="button" data-bs-target="#demo" data-bs-
slide="next">
  <span class="carousel-control-next-icon"></span>
</button>
</div>

```

Bootstrap Carousel / Slideshow

**Left
controls/
icons**



**Right
controls/
icons**

Indicators/dots

Quiz



1. How many columns are allowed in a bootstrap grid system?

a) 10

b) 11

c) 12

d) None of these Above

Ans: c) 12

Quiz



2. Which of the following class in Bootstrap is used to provide a responsive fixed width container?

a).container-fixed

b) .container-fluid

c) .container

d) All the Above

Ans: c) .container

Quiz



3. Which of the following class in Bootstrap is used to create a dropdown menu?

a).dropdown

b) .select

c) .select-list

d) All the Above

Ans: a).dropdown

Quiz



4. Which of the following class in Bootstrap is used to create a large button?

a).btn-xl

b) .btn-lrg

c) .btn-large

d) .btn-lg

Ans: d) .btn-lg

Quiz



5. Say True or False: CSS is more complex than Bootstrap because there is no pre-defined class and design.

a) True

b) False

Ans: a) True

THANK YOU